🤌 XAI: Evaluating Interpretability

FAMAF - UNC

November 3, 2025

🧉 Disclaimer ---

This course is based on

Explainable Artificial Intelligence

From Simple Predictors to Complex Generative Models

Spring 2023, Harvard University

https://interpretable-ml-class.github.io/

Overview ---

- Evaluating interpretability in the interpretable ML community:
  - Interpretability depends on human experience of the model
  - Disagreement about the best way to measure it
- These papers:
  - Evaluating factors related to interpretability through user studies

🧉 Other Relevant Fields ---

- **Human-Computer Interaction (HCI):**
  - Theories for how people **interact with technology**
- **Psychology:**
  - Theories for how people **process information**
- Both have thought carefully about **experimental design**

🧉 Outline ---

- Research paper: **"Human Evaluation of Models Built for Interpretability"** by Lage et al.
- Research paper: **"Manipulating and Measuring Model Interpretability"** by Poursabzi-Sangdeh et al.
- Discussion

🧉 Paper 1 ---

The Seventh AAAI Conference on Human
Computation and Crowdsourcing (HCOMP-19)

Human Evaluation of Models Built for Interpretability

Isaac Lage,[*1] **Emily Chen,**[*1] **Jeffrey He,**[*1] **Menaka Narayanan,**[*1]
Been Kim,[2] **Samuel J. Gershman,**[1] **Finale Doshi-Velez**[1]
[1]Harvard University, [2]Google
isaaclage@g.harvard.edu, {emily-chen, jdhe, menakanarayanan}@college.harvard.edu
beenkim@google.com, gershman@fas.harvard.edu, finale@seas.harvard.edu

Abstract

Recent years have seen a boom in interest in interpretable machine learning systems built on models that can be understood, at least to some degree, by domain experts. However, exactly what kinds of models are truly human-interpretable remains poorly understood. This work advances our understanding of precisely which factors make models inter-

human-simulatable models include human-simulatable regression functions (e.g. Caruana et al., 2015; Ustun and Rudin, 2016), and various logic-based methods (e.g. Wang and Rudin, 2015; Lakkaraju, Bach, and Leskovec, 2016; Singh, Ribeiro, and Guestrin, 2016).

But even types of models designed to be human-simulatable may cease to be so when they are too complex

🧉 Definitions ---

- **Human-simulatability**: The ability of humans to mentally trace through a model's decision-making process and accurately predict its outputs for given inputs, essentially simulating how the model would behave.
- **Decision set**: A logic-based machine learning model consisting of a collection of independent logical rules, where each rule maps a combination of input features to an output prediction.
- **Decision set complexity**: A decision set is more complex when it has more rules, more conditions per rule, or conditions that are more cognitively difficult to evaluate.

Contributions ---

Research Questions:

- Which types of **decision set complexity** most affect human-simulatability?
- Is relationship between complexity and human-simulatability **context dependent**?

Approach:

- Large scale, carefully controlled user studies

Decision Sets ---

- **Logic-based models** are often considered interpretable
- Many approaches for **learning them from data** (Decision Trees)

faithful and cold or **brave and passive** → **candy or dairy** and **fruit**

sleepy or **patient and obedient** → **spices and grains** or **dairy**

brave and **sleepy or patient or laughing** → **dairy and fruit** or **grains**

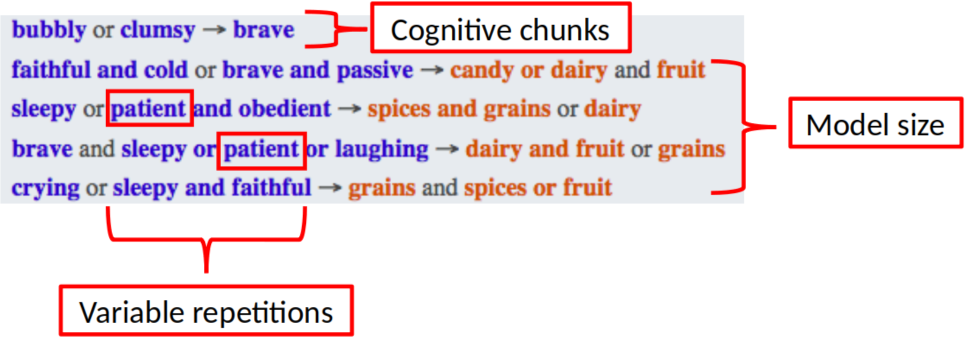
crying or **sleepy and faithful** → **grains** and **spices or fruit**

🧉 Regularizers ---

- **Regularizer (paper context)**: A technique that penalizes certain types of complexity during decision set training (such as number of rules, conditions per rule, or specific types of conditions) to make the models **less complex and more interpretable**.
- What kinds of complexity is it **most urgent to regularize** to learn interpretable models?

🧉 Types of Complexity ---

- **Model size**: Number of rules/lines in the decision set
- **Variable repetitions**: Same variable appearing multiple times
- **Cognitive chunks**: Number of distinct logical conditions



What if we optimized the models with data?

Context: Domains ---

Low Risk: Alien meal recommendation

High Risk: Alien medical prescription

Design Question

What if we used 2 different real domains?

🧉 Context: Tasks ---

- **Simulation:** - What would the model recommend the alien? - **Verification:** - Is *milk and guava* a correct recommendation?
- **Counterfactual:** - If *patient* were replaced with *sleepy*, would the correctness of the *milk and guava* recommendation change?

What if we used more realistic tasks?

🧉 Tradeoff Between Control and Generalizability ---

- Tradeoff between the ability to tightly control the experiment and running it under realistic conditions (generalizability)

Tightly controlled $\longleftrightarrow$ **Realistic**

This paper

🧉 Procedure ---

- Experiment posted on Mturk
- Takes around 20 minutes
- Participants paid 3 USD
- Excluded participants who could not complete practice questions
- Total: 50-70 participants out of 150

Instructions → 3-6 practice questions → 15-18 test questions → Payment code

🧉 Statistical Analysis: Linear Model ---

We use a linear model for each metric in each experiment:

- Response time
- Accuracy
- Satisfaction

Example – Model Size, Response Time:

- **Step 1:** Fit linear regression to predict response time from number of lines and number of output terms
- **Step 2:** Interpret coefficients as effects of number of lines and number of output terms on response time

🧉 Statistical Analysis: Multiple Hypothesis Testing ---

🧉 Results: Complexity Increases Response Time ---

Recipe Domain

🧉 Results: Type of Complexity Matters ---

Response time for: cognitive chunks > model size > repeated terms

🧉 Results: Consistency - Domains, Tasks, Metrics ---

Results consistent across domains, tasks and the response time and subjective difficulty metrics

Results: Counterfactuals Are Hard ---

The counterfactual task is much more challenging than simulation!

🧉 Discussion: Paper 1 ---

- Consistent guidelines for interpretability
- Simplified tasks to measure interpretability
- Using Mturk workers as a proxy for domain experts

Key Takeaway: Different types of complexity affect human-simulatability differently, with cognitive chunks having the strongest effect

🧉 Paper 2 ---

🧉 Manipulating and Measuring Model Interpretability ---

Forough Poursabzi-Sangdeh, Daniel G. Goldstein, Jake M. Hofman
Jennifer Wortman Vaughan, Hanna Wallach

Microsoft Research

🧉 Motivation ---

- Interpretability as a latent property that can be manipulated or measured indirectly
- What are the factors through which it can be manipulated effectively?
- Bring HCI methods to interpretable ML since interpretability is defined by user experience

🧉 Contributions ---

Research Questions:

- How well can people estimate what a model will predict?
- How much do people trust a model's predictions?
- How well can people detect when a model has made a sizable mistake?

Approach:

- Large-scale, pre-registered user studies to answer these questions in the context of linear regression models

🧉 Comparison to Paper 1 ---

- Studies **linear regression models** instead of decision sets
- Measures people's ability to make their **own predictions** in addition to forward simulation
- Uses **real-world housing dataset** and models optimized with data

Ways to Manipulate Interpretability ---

CLEAR-2: 2 features, transparent

BB-2: 2 features, black-box

🧉 Procedure ---

- Participants shown:
  - **Training:** 10 apartments
  - **Testing:** 12 apartments (this is the data they use)
- Participants paid 2.5 USD
- 750-1,250 participants per experiment

Each Trial:

Forward simulate model's prediction $\rightarrow$ View model's true prediction $\rightarrow$ Make own prediction

🧉 Statistical Analysis: Participant Specific Effects ---

A repeated measures experimental design

- Each participant makes many predictions
- Use a **mixed-effects model** to control for correlations between a participant's responses
- Assumes a random, participant-specific effect

🧉 Statistical Analysis: Multiple Hypothesis Testing ---

- **Pre-registering hypotheses** corresponds to deciding and publishing which analyses you will run **before collecting data**
- Reduces the probability that effects were discovered by chance

Example

`https://aspredicted.org/xy5s6.pdf`
"We will use 2-by-2 ANOVA for statistical analysis of the effect of number of features and model clarity on final deviation from model's prediction and simulation error."

🧉 Design Choices ---

- Randomized the order of the first 10 (normal) apartments and fixed the order of the last 2 (unusual)
- All participants are shown an identical set of apartments
- Each participant completed a single condition (between subjects design)

Fix sources of randomness	**Randomize as much as possible**
Can introduce bias	Increases variance

🧉 Results: Simulating Small, Transparent Models ---

Experiment 1: New York City prices

🧉 Results: No Difference in Trust or Prediction ---

Experiment 1: New York City prices

Trust (Deviation)

Prediction Error

Key Finding

🧉 Results: Clear Models Make Mistakes Worse ---

Experiment 1: New York City prices - Apartment 12 (1 bed, 3 bath)

🧉 Additional Experiments ---

Experiment 2: Scaled Prices

- Scaled down prices to better reflect national average
- **Same results** for simulation accuracy

🧉 Additional Experiments (cont.) ---

Experiment 3: Better Trust Metrics

- Scaled down prices to better reflect national average
- Better trust metrics
- **No significant difference in trust between models**

🧉 Additional Experiments (cont.) ---

Experiment 4: Attention Check

- Attention check for unusual features
- **People catch more errors** when explicitly asked to look for unusual inputs

🧉 Discussion: Paper 2 ---

Key findings from the experiments:

- Highlighting weird inputs helps catch errors
- Having people predict before seeing the model helped catch errors
- **Transparency actually makes people worse at catching errors**

Counterintuitive Result

More interpretable models (transparent, fewer features) led to worse error detection on unusual cases

🧉 General Discussion ---

Comparing Both Papers:

- **Paper 1:** Decision sets, controlled synthetic domains, cognitive complexity matters
- **Paper 2:** Linear models, real-world data, transparency can backfire

Key Lessons:

- Interpretability is multifaceted and context-dependent
- User studies are essential for evaluating interpretability
- Simplified models $\neq$ always better decision-making
- Different tasks require different evaluation metrics

Thank You!